

What MC Dropout Actually Does, Step by Step

The Task 6 pipeline drawn and explained: why it loops per fold, what fifty stochastic passes produce, and how two thresholds turn a distribution into a routing decision.

INPUT	FINGERPRINT	NODULES	RUNTIME
5 fold checkpoints + cache	5922c683	322 / 187 patients	~12 min, CPU

1 Where we start from

Tasks 04-01 and 05-01 are complete on the rebuilt v2 cache. Both printed fingerprint 5922c68330c537b8 , so they were scored on the same 322 nodules in the same folds and are properly paired.

	RESNET-18	VIT-TINY	VIT-TINY TUNED
Pooled AUROC	0.8833 [0.845–0.919]	0.6923 [0.626–0.756]	0.5548
Per-fold AUROC	0.9237 ± 0.027	0.6930 ± 0.111	–
Accuracy	78.6%	63.4%	–
Sensitivity / Specificity	0.862 / 0.688	0.779 / 0.447	–
False negatives	25 of 181	40 of 181	–
ECE	0.1372	0.0693	–

Pooled out-of-fold results on the corrected (v2) geometry, 322 nodules from 187 patients.

THE TUNED VIT SETTLES A QUESTION WE HAD LEFT OPEN

The lower learning rate with warmup made ViT-tiny worse — 0.5548 against 0.6923. The hypothesis that the CNN's schedule was handicapping the transformer is therefore **refuted by our own pre-registered experiment**. The ViT's weakness is data scale and architecture, not the optimiser settings.

That is a stronger result for Task 5 than a bare comparison: we tested the obvious objection in advance and can report that it does not hold.

And it decides what the orchestrator may escalate to

Before building an escalation path to the ViT, the question is whether the ViT is right where ResNet is wrong. Run on the v2 out-of-fold predictions:

CANDIDATE	CORRECT ON RESNET'S 69 ERRORS	OWN BASE RATE	GATE
ViT-tiny	34 / 69 = 49.3%	63.4%	FAILS
ViT-tiny tuned	24 / 69 = 34.8%	51.2%	FAILS

Complementarity gate: an alternative specialist is only worth calling if it beats its own base rate on the primary's failures.

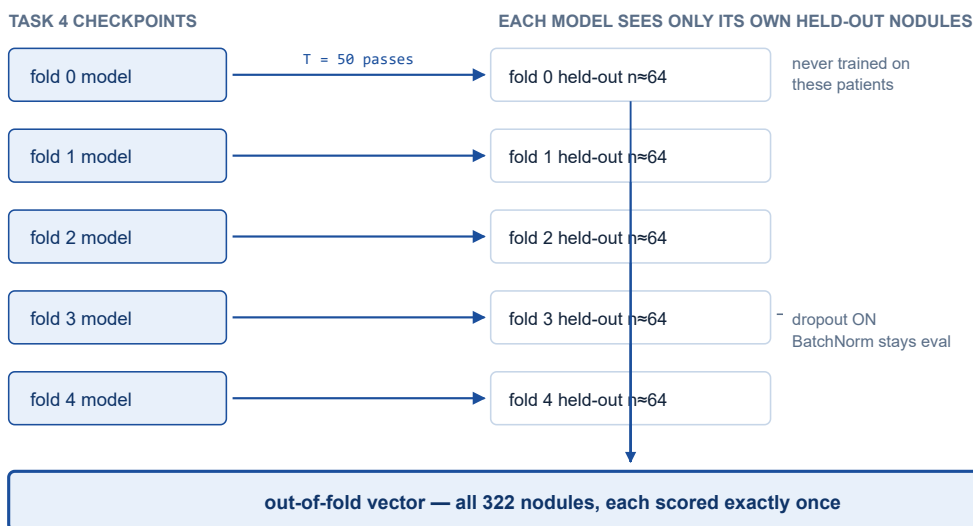
Both fail, and averaging the two models' probabilities *lowers* AUROC to 0.8634 from 0.8833. **The ViT must not be the alternative specialist.** Where ResNet struggles, ViT struggles more.

ONE WEAK COUNTER-SIGNAL, RECORDED BUT NOT ACTED ON

On the 25 nodules ResNet actually *missed as cancers*, ViT was right on 17 (68%), slightly above its 63.4% base rate. On 25 cases that difference is about one nodule's worth and is not significant. It is noted because false negatives are the clinical target, and because it is worth re-testing once the dataset is larger — not because it justifies building the branch now.

2 The fold loop — and why it exists

Task 4 did not produce one model. It produced **five**, one per cross-validation fold, each trained on four fifths of the patients. There is no single "final" model to sample.



The same loop also samples each fold's validation slice, used only to calibrate thresholds in step 5.

Why the loop, not one model over everything. A model is confident on data it trained on, so running any single fold model across all 322 nodules would measure confidence on memorised cases. Each nodule must be sampled by the one model that never saw its patient — the same out-of-fold discipline as Task 4, applied to uncertainty.

3 What fifty passes produce

A normal prediction is one number. Keeping dropout switched on and running the same patch fifty times gives a *distribution* of fifty numbers — and the shape of that distribution is the signal.

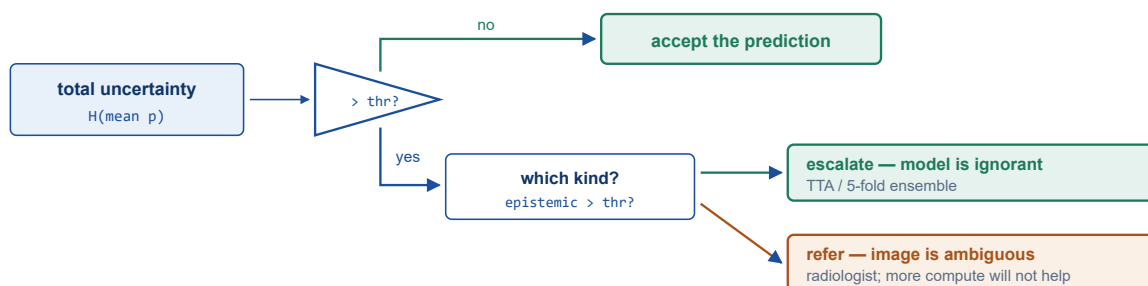


Reading the spread:

tight cluster → every masked version of the network agrees → low epistemic
wide scatter → the answer depends on which units were dropped → high epistemic
all fifty sitting near 0.5 → the network is consistently undecided → high aleatoric
Entropies are divided by $\ln 2$, so every number lands in $[0, 1]$.

The split is the point. A single confidence number cannot tell "the model is out of its depth" apart from "this lesion is genuinely borderline", but those need opposite responses. Note the arithmetic: *aleatoric* is the average of the individual entropies, *total* is the entropy of the average, and epistemic is what is left over.

4 Two thresholds, three outcomes



Both thresholds are calibrated on the **validation** slices only — never on the out-of-fold set the results are reported from.
This is the whole decision. Everything above it exists to make these two numbers trustworthy.

The routing rule in full. One threshold decides *whether* to act, the second decides *what* to do. Sending an aleatoric-dominated case to another model burns compute on a lesion no model can resolve; sending an epistemic-dominated case to a human wastes a radiologist on something a better model would have got.

5 The implementation, step by step

1 Load the cache and the five checkpoints, then verify

Each checkpoint stores the fingerprint it was trained under. If it does not match this session's data the run stops — a mismatched fold partition would mean the "held-out" nodules include ones the model trained on, and the uncertainty would look good and mean nothing.

2 For each fold: one deterministic pass, then fifty stochastic ones

The deterministic pass reproduces the Task 4 baseline prediction. Then `enable_mc_dropout()` puts the model in eval mode and re-enables *only* the dropout layers — `model.train()` would also switch BatchNorm back on, making predictions depend on which other nodules share the batch.

3 Turn fifty numbers into four

Mean probability, total entropy, aleatoric, epistemic — figure 2. Written into full-dataset vectors so every nodule ends up with a deterministic probability, an MC probability and its decomposition, all out-of-fold.

4 Report the calibration — the actual deliverable

ECE and Brier, deterministic against MC-averaged, per fold and pooled, with patient-clustered confidence intervals. AUROC alongside as a control: averaging should improve calibration without costing discrimination. ResNet's current ECE is 0.1372, so there is real room to improve.

5 Calibrate the thresholds on validation only

The per-fold validation slices — sampled in step 2 and untouched since — set the referral and epistemic thresholds. Choosing them on the out-of-fold set would invalidate every number in step 4.

6 Prove the uncertainty is worth acting on

Three checks, each able to fail: does uncertainty rank the model's own errors above chance; does the risk-coverage curve beat *random* referral; and does the aleatoric term correlate with the radiologists' own disagreement more strongly than the epistemic term does. The third is the one no published work appears to have run on LIDC.

7 Emit the contract

`uq_interface_contract.json` plus per-nodule uncertainty. This is the boundary: Task 6 ends here, and the Qwen3 orchestrator begins by reading this file.

WHAT THIS MEANS FOR THE ESCALATION BRANCH

Figure 3's "escalate" box should be filled with **TTA and the 5-fold ensemble**, not the ViT. The complementarity gate in section 1 was run before building anything and both ViT variants failed it. Reporting that test — and that we declined to build the branch because of it — is better methodology than quietly wiring up a box because the diagram had one.